Jackson Mock

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EDUCATION

North Carolina State University Raleigh, NC
M.S. in Computer Science (Accelerated B.S./M.S. Program) | GPA: 3.9/4.0 *Expected May 2027*
B.S. in Computer Science, Minor in Business Administration | GPA: 3.8/4.0 *May 2026*

Coursework: Applied Web-Based Client-Server Computing, Foundations of Cryptography, Cloud Architecture, Concepts and Facilities of Operating Systems, Automated Learning and Data Analysis, Human Computer Interaction

PROFESSIONAL EXPERIENCE

Web Developer | SkySouth | Burlington, NC | Part-Time *December 2025 - March 2026*

- Built a custom PHP theme with Tailwind CSS to replace SkySouth's legacy site, owning architecture, UI/UX, and responsive web implementation.
- Increased active users by 652% and organic search sessions by 1,097% YoY by implementing SEO-focused site architecture, metadata, and content targeting high-intent search queries.
- Implemented page-specific Schema.org JSON-LD structured data and a 301 redirect map to preserve search rankings through the site migration.

Software Engineer Intern | Pearson | Durham, NC *June 2025 - August 2025*

- Saved over 100,000 hours of manual labor annually by developing a Python image processing pipeline to automatically validate millions of exams.
- Cut the false positive rate from 53.5% to 1.9% on a 1,000-exam validation sample by designing and iteratively refining OCR error correction logic.
- Extracted exam data from scanned pages using EasyOCR for text recognition and OpenCV for image preprocessing.
- Made multi-hour batch runs reliable with multithreading, checkpoint-based crash recovery, and SMTP status reports.

Research Assistant | Wolfpack Innovations Software Engineering Research | Raleigh, NC *June 2024 - September 2024*

- Automated differential testing across 25 C compiler and optimization-level configurations to detect miscompilations where identical source code produced inconsistent output.
- Wrote reproducible bug reports with minimal test cases and compiler flags, enabling developers to isolate the root cause.
- Developed web scraping applications to extract data and test files from hundreds of web pages.

PROJECTS

Hobbs | TypeScript, Node.js, Fastify, PostgreSQL, PostGIS, Docker, React | Personal *October 2026 - Present*

- Building a full-stack logbook and social app for student pilots: log multi-leg flights, track progress toward FAA private pilot requirements, map routes automatically with ADS-B data, and share flights with friends.
- Designed a PostgreSQL schema for users, flights, multi-leg routes and aircraft, with database constraints that enforce logbook rules and exact decimal flight time, managed through reversible SQL migrations.
- Loaded 16,000+ U.S. airports into PostGIS geography with a GiST index for nearest-airport and distance queries.
- Built a typed Fastify REST API that shares Zod validation schemas with the React client.

3D Flight Map Visualizer | React, Tailwind CSS, Deck.gl, MapLibre | Personal *September 2025 - October 2025*

- Built an interactive 3D web app animating an airline's 22,000+ flights over 23 years as arcs from origin to destination.
- Parsed flight logs for ICAO airport codes and generated monthly JSON datasets of coordinates.
- Kept rendering smooth by animating only a configurable sliding window of arcs, collapsing expired arcs into static endpoint dots that preserve the flight history.

SKILLS

Languages: Java, C, Python, HTML, CSS, JavaScript, TypeScript, Assembly, SQL, PHP
Frameworks: React, Spring, Tailwind CSS, Deck.gl, Fastify, Express.js
Databases: PostgreSQL, PostGIS, Supabase, MongoDB, MariaDB
Technical Skills: RESTful Design, Unit Testing, Git, Docker, Figma, OpenCV, EasyOCR